# RUDRA PRATAP SINGH

**AI/ML and Full-Stack Developer | Generative AI, AI Agents, RAG, LLM Evaluation and AI Security**

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Portfolio: rudraksh01-tech.github.io

## SUMMARY

Computer Science undergraduate (VTU, CGPA 8.0) building Generative AI and Large Language Model (LLM) applications: AI agents, RAG, LLM evaluation and AI security. Shipped three live, tested projects (Python, LangGraph, FastAPI, React, PostgreSQL, Docker, Kubernetes); Founding Software Engineer Intern at FERMOR; seeking AI and Machine Learning (AI/ML) internships.

## EDUCATION

**Bachelor of Engineering (B.E.) in Computer Science and Engineering (CSE)** 2024 - 2028 (Expected) | 5th semester
Cambridge Institute of Technology, North Campus, Bengaluru | VTU Current CGPA: 8.0 / 10

## EXPERIENCE

**Founding Software Engineer Intern**, FERMOR | *Bengaluru, Karnataka (Hybrid)* Sep 2026 - Oct 2026

- Redesigned key website pages end to end with a cleaner frosted-glass UI/UX, and optimised page structure, headings, metadata and content for Search Engine Optimization (SEO).
- Drove **about 30x** the usual daily traffic on launch day (a month's worth in one day) through the redesign and SEO work; used web analytics to measure the impact and plan next steps.

## PROJECTS

**Dwarpal** | AI Security Gateway for LLM Agents (MCP) rudraksh01-tech.github.io/dwarpal
*Python, LangGraph, MCP, FastAPI, PostgreSQL, pgvector, Redis, Docker, Kubernetes, Helm | github.com/rudraksh01-tech/dwarpal*

- Built an open-source AI security gateway (guardrails) between LLM agents and MCP tool servers that blocks prompt injection, tool poisoning, rug pulls and sensitive-data leaks in English, Hindi, Hinglish and Kannada.
- Architected a LangGraph policy graph with human-in-the-loop approval: SHA-256 tool pinning, cross-server data-flow (taint) tracking, multilingual rules, Llama Prompt Guard 2 and an LLM-as-a-judge, with Redis-cached verdicts.
- Cut data leaks from **5 of 25** attack runs to **0 of 25** with **0 false blocks** over 60 repeated agent runs; showed that Prompt Guard 2 scores a politely worded attack **0.001** (safe).
- Stored tool pins and audit logs in PostgreSQL with pgvector; shipped Docker images and a Helm chart that CI/CD (GitHub Actions) deploys and tests on Kubernetes; **34** unit and integration tests.

**AyurIP-Sahayak** | Multilingual RAG Assistant ayur-ip-sahayak.vercel.app
*Python, FastAPI, React, Google Gemini API, NumPy, Docker, GitHub Actions | github.com/rudraksh01-tech/AyurIP-Sahayak*

- Launched a Retrieval-Augmented Generation (RAG) assistant for Ayurveda and Indian IP law that answers in English, Hindi, Kannada and Hinglish with page-level citations and **under 2 s** latency to the first streamed words.
- Engineered retrieval from scratch (PDF ingestion, section-aware chunking, Gemini embeddings, vector search, BM25 + Reciprocal Rank Fusion hybrid search): Hit@1 **0.95** and MRR **0.97** on 38 labelled questions vs 0.47 and 0.60 for BM25.
- Validated answers with an LLM-as-a-judge: **0.99** faithfulness (166 of 167 claims), 0.98 citation precision, **15 of 15** planted errors caught; fixed out-of-memory deploys by replacing a local PyTorch model with an embeddings API; 47 tests.

**NestFinder** | Agentic AI Assistant for Flat Search nestfinder-woad.vercel.app
*Node.js, Express.js, React, MongoDB, Groq API, Langfuse, OpenTelemetry | github.com/rudraksh01-tech/nestfinder*

- Developed a tool-calling (function calling) AI agent from scratch: 5 tools, streaming, structured JSON outputs, a repair loop for invalid tool calls and automatic model fallback on API rate limits.
- Raised tool-choice accuracy from **18/20 to 20/20** with a 20-case agent evaluation (Hindi, Hinglish and prompt-injection inputs) that exposed a schema bug.
- Instrumented LLM observability with OpenTelemetry and Langfuse (PII masked); added JWT authentication, rate limiting and MongoDB geospatial queries; deployed on Vercel and Render with CI.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, Java, SQL | **Frontend:** React, Vite, Tailwind CSS, HTML, CSS

**Generative AI and LLMs:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Agentic AI, LangGraph, Model Context Protocol (MCP), Tool Calling (Function Calling), Prompt Engineering, Structured Outputs (JSON Schema), Embeddings, Natural Language Processing (NLP), Google Gemini API, Groq API

**Retrieval and Vector Search:** Vector Databases (pgvector), Vector Search, Semantic Search, Hybrid Search (BM25, RRF), Reciprocal Rank Fusion, NumPy

**LLM Evaluation and AI Security:** LLM Evaluation, LLM-as-a-Judge, Guardrails, Prompt Injection Defense, OWASP Top 10 for LLMs, LLM Observability

**Backend and Databases:** FastAPI, Node.js, Express.js, REST APIs, Async Python, Pydantic, PostgreSQL, Redis, MongoDB, JWT

**DevOps and Cloud:** Docker, Kubernetes, Helm, CI/CD (GitHub Actions), Git, Linux, Vercel, Render, Unit Testing (pytest)

**Core CS:** Data Structures and Algorithms, Object-Oriented Programming (OOP), Operating Systems, DBMS, Computer Networks